BAO, Conglin

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EDUCATION

Emory University
Master of Science in Public Health, Concentration on Biostatistics Aug 2024 – May 2026
Key Modules: Probability Theory 1 (A), Biostatistical Methods (A), Statistical Inference I (A-), Applied Linear Models (A-), Modern Regression Analysis (A)
Data Science Certificate Coursework: Machine Learning (A), R Programming (A), Python Programming (A), Data Science Toolkit (A)
Honors: GPA: 3.908/4.00; Poster Presentation Award at Georgia Statistics Day 2025

University of Nottingham Ningbo China
Bachelor of Science (Hons) in International Economics and Trade Sep 2020 - Jul 2024
Key Modules: Microeconomic Theory, Macroeconomic Theory, Econometrics, Development Economics, Python Programming for Economics
Honors: GPA: 3.62/4.00 (Top 5% in Junior Year); 2022-2023 Provost's Scholarship (5%)

Summer School in University of Nottingham Ningbo China
Mathematical Economics and Financial Econometrics Jul 2022 - Aug 2022
Key Modules: Linear Algebra with MATLAB, Probability Theory, Calculus and Optimization, R Programming for Economics, R for Finance, Expectations and Learning in Macroeconomics and Finance
Honors: Excellent (15%)

PUBLICATIONS & MANUSCRIPTS

- **Bao, C.***, Dai, Q.*, & Yang, J. (2025). *Cross-Validated Deep Learning Models for Inferring Alzheimer's Pathology Trajectories from Longitudinal Clinical Measures. (* indicates co-first authorship)*
  - Manuscript in preparation (Draft completed; available upon request).
  - Targeted for submission to ***Alzheimer's & Dementia***.
  - Poster presented at **Georgia Statistics Day 2025**, University of Georgia, Athens, GA.
  - *Reproducible code available at:* github.com/RichardFranklinBAO/AD_Pathology_Trajectory_DL
- Warner S, **Bao C**, He Y, Wiltse Nicely K, Carlson N, Chandler R, & Manyam R. (2025). *Leveraging Machine Learning to Explore Anesthesia Exposure and Severe Maternal Morbidity in Nulliparous Women: A Secondary Analysis of nuMoM2b.*
  - In Review at ***The British Journal of Anaesthesia***.
- Manyam, R., **Bao, C.,** He, Y., Zhang, Y., & Keeling, B. (2025). *Interpretable Artificial Intelligence Risk Model to Predict Readmission after Coronary Artery Bypass Grafting (CABG).*
  - Abstract accepted for Rapid Fire Oral presentation at the ***106th Annual Meeting of the American Association for Thoracic Surgery (AATS)***, Chicago, IL (May 3, 2026).
  - Methods: recursive feature elimination (RFECV), SHAP, and clinical electronic health records (EHR) modeling.
- Manyam, R., He, Y., **Bao, C.,** Gupta, P., & Zhang, Y. (2025). *Interpretable AI-enabled Risk Model to Predict Pneumonia after Coronary Artery Bypass Grafting.*
  - Abstract submitted to the ***Eastern North American Region (ENAR), International Biometric Society, 2026 Spring Meeting***, Indianapolis, IN.
  - Methods: Random Forest, CatBoost, SHAP; dataset of 13,650 CABG patients.

TECHNICAL SKILLS

- **Mathematical Proficiency:**
  - Multivariate Calculus, Linear Algebra (Matrix Factorization, Eigenvalues), Probability Theory (calculus based), Constrained Optimization (KKT, Lagrangians), Mathematical Statistics (Probability-based Inference). (See Appendix for detailed coursework mapping)
- **Statistical Programming & Languages:**
  - **Python:** Advanced. Developed **custom Scikit-learn estimators** (via BaseEstimator) and reproducible Deep Learning pipelines. Core Libraries: NumPy, Pandas, Matplotlib, SciPy.
  - **R:** Advanced. Specialized in reproducible workflows and package management. Libraries: Tidyverse (dplyr, ggplot2), here, config.
  - **Bash/Shell:** Proficient in command-line navigation, shell scripting, and automating file operations in Unix/Linux environments.
  - **SAS & Stata:** Proficient in data management and regression analysis for public health datasets.
  - **SQL:** Experience in querying and managing structured databases.
- **Machine Learning & Deep Learning:**
  - **Advanced ML/DL: PyTorch** (Expertise in building LSTM/BiLSTM architectures, utilizing nn.utils.rnn for **variable-length sequence packing/padding**, and TorchMetrics for modular evaluation).
  - **Algorithms:** Scikit-learn (Custom pipeline integration), XGBoost, CatBoost. Implemented Recursive Feature Elimination (RFECV) and KNN imputation.
  - **Explainable AI (XAI): SHAP**. Proficient in generating Beeswarm, Force, and Waterfall plots for model interpretability.
- **Reproducible Research & DevOps (Advanced):**
  - **Automation: GNU Make**. Architected fully reproducible workflows using Makefiles to automate script conversion (e.g., **.ipynb to .py**, **.Rmd to .R**) and orchestrate end-to-end data pipelines.
  - **Containerization: Docker**. Experienced in writing Dockerfiles, building images, and using DockerHub to ensure environment consistency across platforms.
  - **Dependency Management: renv**. Skilled in initializing project libraries, managing lockfiles, and restoring consistent R environments.
  - **Version Control: Git & GitHub**. Advanced workflows including branching strategies, pull requests (PRs), resolving merge conflicts, and remote repository management.
- **Scientific Reporting:**
  - **R Markdown:** Expertise in creating **parameterized reports** for dynamic output generation and integrating **LaTeX** for mathematical notation.

RESEARCH EXPERIENCE

Leveraging Machine Learning to Explore Anesthesia Exposure and Severe Maternal Morbidity in Nulliparous Women: A Secondary Analysis of nuMoM2b.

Research Assistant, Department of Biostatistics and Bioinformatics, Emory University, Atlanta, GA, USA *Directed by Professor Ramesh Manyam* *Sep 2025 – Dec 2025*

- Built a reproducible pipeline that partitions a 9,289×11,633 wide table into 24 datasets by prefix, applying custom schema rules to standardize demographic and pregnancy outcome features, ensuring individual ID normalization and consistent exports (Python/R).
- Developed a causal inference framework on the nuMoM2b cohort (N=8,056) to estimate the ATT of general vs. neuraxial anesthesia.

- Addressed extreme class imbalance (1:141) using Propensity Score Matching with 1:1 nearest-neighbor caliper matching; reduced maximum SMD from 0.99 to 0.21.
- Compared Logistic Regression, Random Forest, and XGBoost for propensity score estimation; selected Logistic Regression to ensure sufficient common support against polarized tree-based distributions.
- Conducted sensitivity analyses (Overlap Weighting) and subgroup evaluation, identifying BMI as a meaningful effect modifier and quantifying an adjusted risk difference of 18.5%.

Interpretable Artificial Intelligence Risk Model to Predict Readmission after Coronary Artery Bypass Grafting

Research Assistant, Department of Biostatistics and Bioinformatics, Emory University, Atlanta, GA, USA *Directed by Professor Ramesh Manyam* *Sep 2025 – Nov 2025*

- Preprocessed and cleaned the CABG dataset by preserving continuous variables, encoding categorical variables into ordered numeric values, and imputing missing data (<10%) using K-Nearest Neighbors (KNN) to ensure model-ready, high-quality inputs for training.
- Developed an explainable machine-learning pipeline using Python (scikit-learn, XGBoost, SHAP, matplotlib) to predict 30-day hospital readmission risk after Coronary Artery Bypass Graft (CABG) surgery based on a dataset of 5,056 patients with 36 clinical variables.
- Applied Recursive Feature Elimination with Cross-Validation (RFECV) to identify the optimal subset of predictive variables, maximizing the F1-weighted score while controlling model complexity and overfitting.
- Generated multi-level SHAP visualizations to interpret model predictions:
  - Bar plots quantified global feature importance, highlighting key predictors such as length of stay, comorbidity index, and anesthesia type.
  - Beeswarm plots illustrated how feature values and directions influenced readmission risk across the patient population.
  - Force and Waterfall plots explained individual predictions, clarifying why specific patients were classified as high or low risk.
- Compared interpretability results across Decision Tree, XGBoost and Random Forest models to ensure robustness and consistency of key predictors.
- Automated feature selection and interpretability report generation, improving pipeline reproducibility and supporting collaboration with clinical researchers.

Cross-Validated Deep Learning Models for Inferring Alzheimer's Pathology Trajectories from Longitudinal Clinical Measures

Research Assistant, Department of Biostatistics and Bioinformatics, Emory University, Atlanta, GA, USA *Directed by Professor Jingjing Yang* *Sep 2024 – Oct 2025*

- Performed rigorous data preprocessing and harmonization on >20,000 longitudinal clinical records to ensure statistical integrity.
- Built a 5-fold cross-validation pipeline to train and evaluate LSTM-based neural networks (LSTM, Bidirectional LSTM (BiLSTM), ReLU variants) on >20 K longitudinal clinical records from decedent participants.
- Used mean out-of-fold R^2 performance to select optimized model architecture, target selection, and hyperparameters for each pathology.
- Generated full-time-step "clean imputations" of AD pathologies for 1.3 K decedents using a 5-fold cross-validation framework, and retrained optimal models on the full dataset to obtain final network weights.
- Applied final models to 800 + living participants to infer longitudinal pathology trajectories, enabling disease-subtype discovery through K-means clustering.
- *Reproducible code and analysis pipeline available at:* github.com/RichardFranklinBAO/AD_Pathology_Trajectory_DL

Intergenerational Educational Mobility: How does parents' education affect children's outcomes by gender

Research Assistant, School of Economics, UNNC, Ningbo, Zhejiang, China *Directed by Professor Sarah Cook* *Sep 2023 - May 2024*

- Managed data collection and cleaning using Stata according to the variables required for use.
- Significantly reduced the dataset from 51,291 entries to 12,862 usable data points facilitating future use.
- Constructed linear and logistic regression models using Stata software and tested the confidence intervals of the parameters to ensure the conclusions are statistically significant and reliable.

The Relationship between the Subjective Well-being and the Income among Parents in China

Research Assistant, School of Economics, UNNC, Ningbo, Zhejiang, China *Directed by Professor Sinchan Mitra* *Nov 2022 - Dec 2022*

- Researched and compiled thesis data, compared it with online survey results (267 responses) and street interviews (38 responses) to validate research conclusions.

The Effect of Inequality on Teamwork

Research Assistant, School of Economics, UNNC, Ningbo, Zhejiang, China *Directed by Professor Tom Lane* *Apr 2022 - May 2022*

- Proposed suggestions for experimental improvements to more accurately simulate the impact of emotional factors on subsequent economic decisions.
- Offered potential interpretations of the data based on experimental outcomes and submitted them for approval to the advisor.

SELECTED ACADEMIC PROJECTS

Regression Modeling of Wage Determinants in U.S. Workers

Course Project, BIOS 526: End-to-End Regression Workflow, Emory University, Atlanta, GA, USA *Fall 2025*

- Conducted Exploratory Data Analysis (EDA) on a cross-sectional dataset of ~3,000 U.S. male workers (2003–2009), assessing wage skewness, outliers, and potential age–wage nonlinearity across education groups.
- Fit a baseline multiple linear regression model for population-level $E[\text{wage} \mid \text{age,education,year,race}]$, and evaluated adequacy using residuals-vs-fitted, residuals vs key predictors and Q–Q diagnostics.
- Extended the model using Generalized Additive Models (GAMs) in R (mgcv) with penalized splines for age and education-specific smooth deviations; selected final specification via AIC and REML-based smoothing.
- Summarized substantive findings with adjusted prediction curves showing nonlinear wage–age trajectories that differ by education; interpreted heavy-tail departures cautiously and emphasized robust inference.

Racial Differences in Glucose Control Among Adults with Type 2 Diabetes

Course Project, BIOS 509: Applied Regression Models, Emory University, Atlanta, GA, USA *Spring 2025*

- Analyzed clinical data from 487 adults with Type 2 Diabetes using **SAS**, comparing fasting glucose levels across racial groups via **one-way Analysis of Variance** (ANOVA) and **Scheffe post-hoc contrasts**.
- **Developed nested logistic regression models** to estimate the probability of high glucose (>150 mg/dL) using weight and hemoglobin A1c (HbA1c), evaluating changes in model fit and race effects using **likelihood ratio tests**.
- Performed model diagnostic checks, including the **Shapiro-Wilk test** for normality and influence analyses (**Cook's Distance, DFFITS, DFBETAS**) to detect impactful observations.
- Interpreted regression coefficients and **odds ratios** to summarize predictor effects and model-based evidence.

Statistical Evaluation of Radial Keratotomy (PERK) Clinical Trial Data

Course Project, BIOS 508: Biostatistical Methods, Emory University, Atlanta, GA, USA *Fall 2024*

- Analyzed clinical trial data from the PERK study (N=386) using **SAS/R** to assess surgical outcomes, generating descriptive statistics and visualizations for postoperative refractive changes.
- Built **univariable linear regression models** to evaluate associations between refractive change and patient characteristics (age, IOP, gender, clear-zone diameter), interpreting **beta estimates** and **R squared** values.
- Applied **Chi-square tests** and **Logistic Regression** to identify predictors of achieving a clinically "good" outcome (-1 to +1 diopters), quantifying effect sizes via **Odds Ratios** and 95% confidence intervals.
- Authored a structured statistical report following **medical journal standards**, integrating diagnostics, methods, and clinical interpretation to support evidence-based conclusions.

ADDITIONAL INFORMATION

- Academic Editing: Proficient in manuscript formatting and submission standards for top-tier journals (e.g., Frontiers in Bioinformatics), ensuring strict adherence to style guidelines
- Language: Mandarin (native), English (proficient), GRE (Full Mark in Math Part)
- Certificate: Junior Mountain Outdoor Instructor by China Mountaineering Association, Sports Leadership by UK Coaching
- Publication: Author of Ancient Poetry for Future, ISBN 978-7-204-16311-3

LEADERSHIP

Touch Rugby Sep 2021 - Dec 2022

- Actively participated in training sessions, boosted team morale; organized pre-game speeches, solicited feedback, and determined optimal player positions to instill a winning mindset in the team.
- Encouraged players to engage in more active communication during injury time.
- Conducted conversations with players who lost confidence due to match mistakes, helping them regain self-assurance for improved performance in subsequent games.

I-Week Leader Aug 2021 - Sep 2021

- Served as a team leader for the 2021 freshman orientation team at the University of Nottingham Ningbo China, dedicating over 136 hours of volunteer service.
- Assisted 12 new students with orientation, greeted them at the school gate, guided them to dormitories, facilitated dorm check-ins, and helped them acclimate to the college environment.
- Organized activities for new students, including watching the film "The Tomorrow War," touring scenic spots in Ningbo (Tianyi Pavilion, Dongqian Lake), and accompanying them to mandatory school health checks.

Sports Leadership Member Jun 2021

- Developed and successfully implemented an original team-building game through a physical education course, enhancing communication and cooperation among team members.
- This experience cultivated my professional ability to lead and instruct diverse groups based on participants' capabilities and course objectives.

Hult Prize Member Oct 2020 - Dec 2020

- Proactively assisted the team leader in organizing brainstorming sessions, offering constructive suggestions that were adopted, and contributed to winning the fourth place in the university-wide competition.

Peer Listener Member Oct 2020 - Nov 2020

- Listened to three peers discuss their lives and pressures; helped alleviate the stress of a number of students through verbal communication therapy.
- Assisted professionals in determining the necessity for further treatment.

APPENDIX: MATHEMATICAL PREPARATION

Calculus (Multivariate & Optimization)

- **ECON2047: Intermediate Mathematical Economics** (Dynamic optimization, continuous-time differential equations, Calculus of Variations, Euler–Lagrange equations, Pontryagin maximum principle)
- **ECON1049: Mathematical Economics and Econometrics** (Multivariable calculus, partial differentiation, constrained optimization, Hessian matrices, KKT conditions, Lagrangian multipliers)
- **CELEN089: Foundation Maths for Economics II** (Differentiation rigor: Taylor & Maclaurin series, L'Hôpital's rule, Implicit differentiation, differential equation setup)
- **CELEN088: Foundation Maths for Economics I** (Calculus foundations: Limits, continuity, critical point analysis, optimization formulation)
- **Summer School: Calculus and Optimization** (Taylor/Maclaurin expansion, limits, L'Hôpital's rule)

Linear Algebra

- **BIOS 526: Modern Regression Analysis** (Linear algebra for high-dimensional data: Eigensystem analysis for PCA, orthogonal projections, matrix operations for spline basis expansion)
- **BIOS 509: Applied Regression Models** (Matrix algebra for linear models: Quadratic forms, Projection matrices, Least squares solution, Rank & Idempotent matrices)
- **ECON1049 & CELEN088** (Matrix Theory: Gaussian elimination, LU Factorization, Matrix inversion conditions, Diagonalization, Definiteness of matrices, Linear transformations)
- **Summer School: Linear Algebra** (Gaussian elimination, LU factorization, vector spaces, separating hyperplane theorem, eigenvalues/eigenvectors)

Probability Theory (Calculus-based)

- **BIOS 512: Probability Theory I** (Rigorous calculus-based probability on σ-algebras, Conditional probability, Moment Generating Functions (MGF), Joint/Conditional distributions, Central Limit Theorem, Law of Large Numbers)
- **Summer School: Probability Theory** (Borel sets, Bayes' theorem, Inclusion–exclusion principle, Convergence of random variables)
- **ECON2033: Econometrics II** (Stochastic processes, Stationarity conditions (Weak/Covariance Stationarity), Integration & Cointegration, Asymptotic convergence concepts)

Mathematical Statistics (Probability-based Inference)

- **BIOS 513: Statistical Inference I** (Rigorous frequentist & Bayesian theory: Likelihood Principle, Sufficiency & Factorization Theorem, Optimality Theory (Rao-Blackwell, Lehmann-Scheffé, MVUE), Fisher Information & Cramér-Rao Lower Bound, Asymptotic Efficiency, Neyman-Pearson Lemma, UMP Tests, Bayesian Inference (Conjugate Priors, HPD Intervals, Empirical Bayes))
- **BIOS 526: Modern Regression Analysis** (Advanced modeling: Penalized Splines, Basis-spline formulation, Bias–Variance tradeoff, Generalized Additive Models (GAMs), Principal Component Analysis (PCA) for dimension reduction, condition indices)
- **BIOS 508: Biostatistical Methods** (Maximum Likelihood Estimation (MLE), log-likelihood optimization, sampling distributions, exact tests)
- **ECON3076: Topics in Time Series Econometrics** (MLE derivation for time-series, Score vector & Hessian matrix construction, Likelihood Ratio/Wald/LM tests)
- **ECON2032: Econometrics I** (Least-squares objective function, Gauss-Markov assumptions, Classical inference tools (t-tests, F-tests, Confidence Intervals), Multicollinearity & Variance Inflation Factor (VIF) diagnostics)
- **ECON3090: Microeconometrics Methods** (Causal identification: Instrumental Variables (IV), Regression Discontinuity (RDD) local treatment effects, Difference-in-Differences (DID), selection bias adjustment)